SDITO Lab PC-CS6G4

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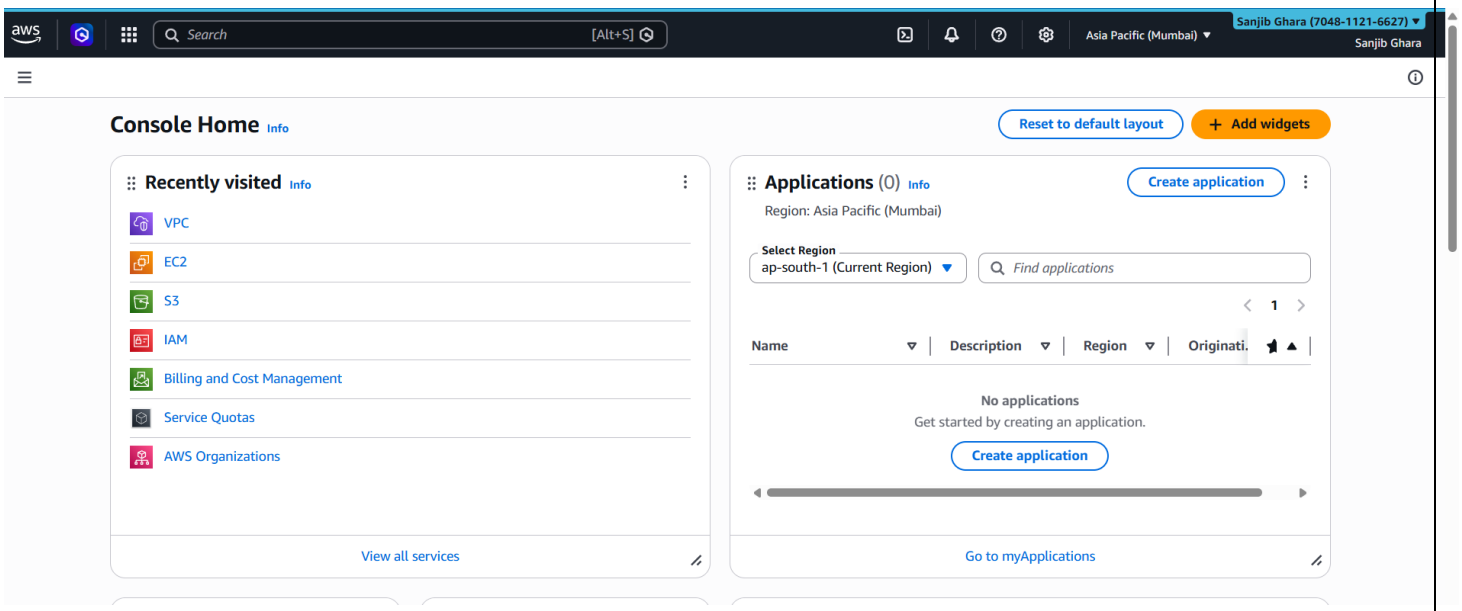
Assignment 14

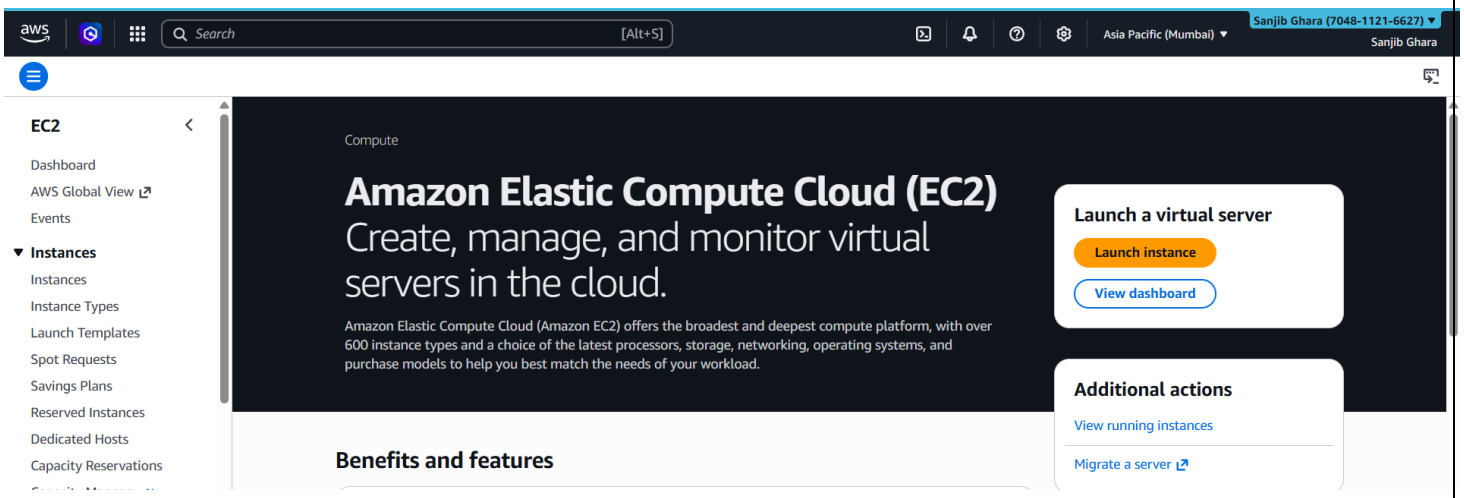
Problem Statement : Create an Elastic IP for an Instance.

Solution:

STEP 1: Login to AWS Console

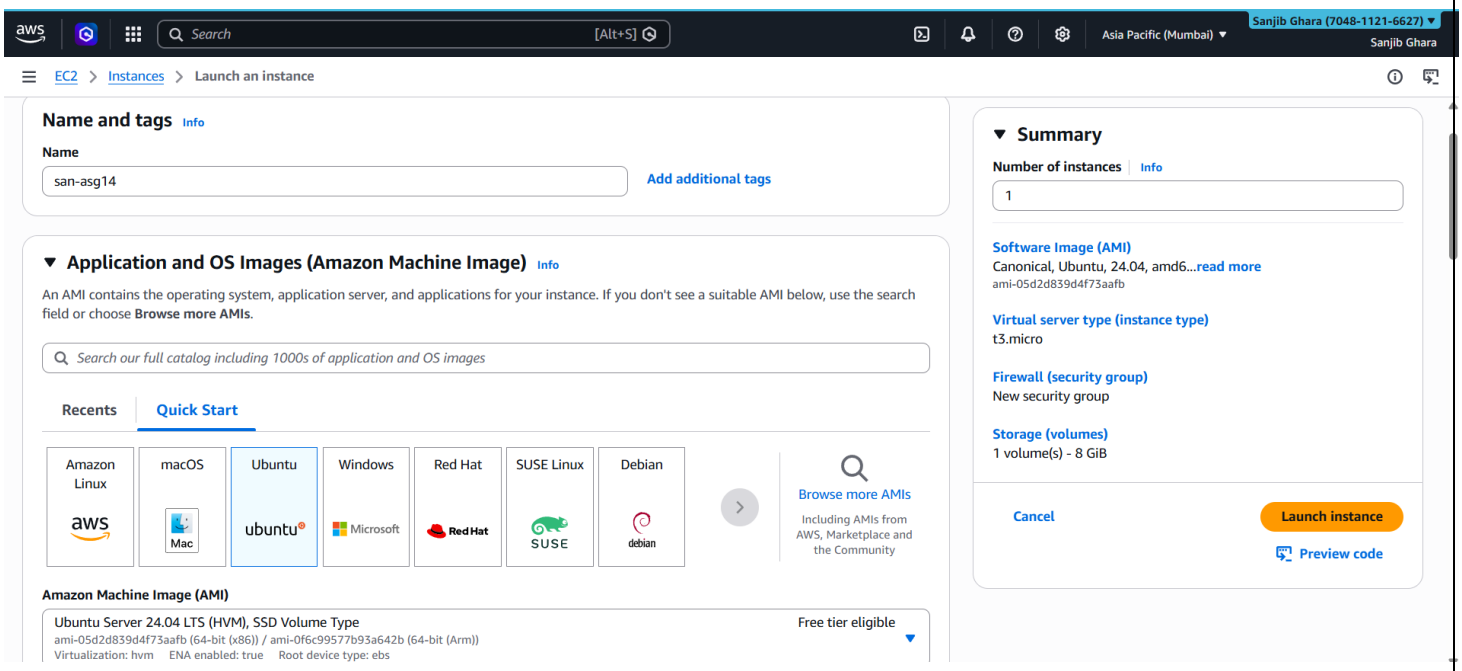


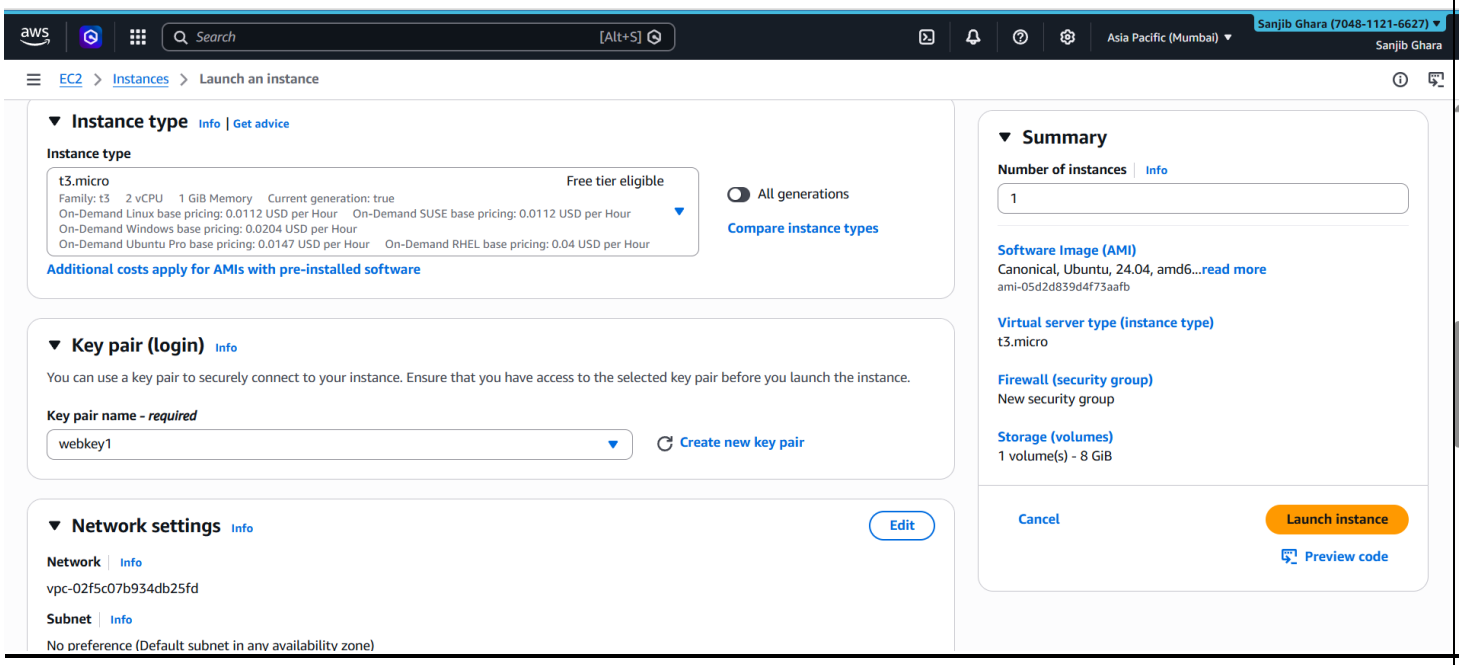
1. Go to EC2 Dashboard
2. Click launch instance



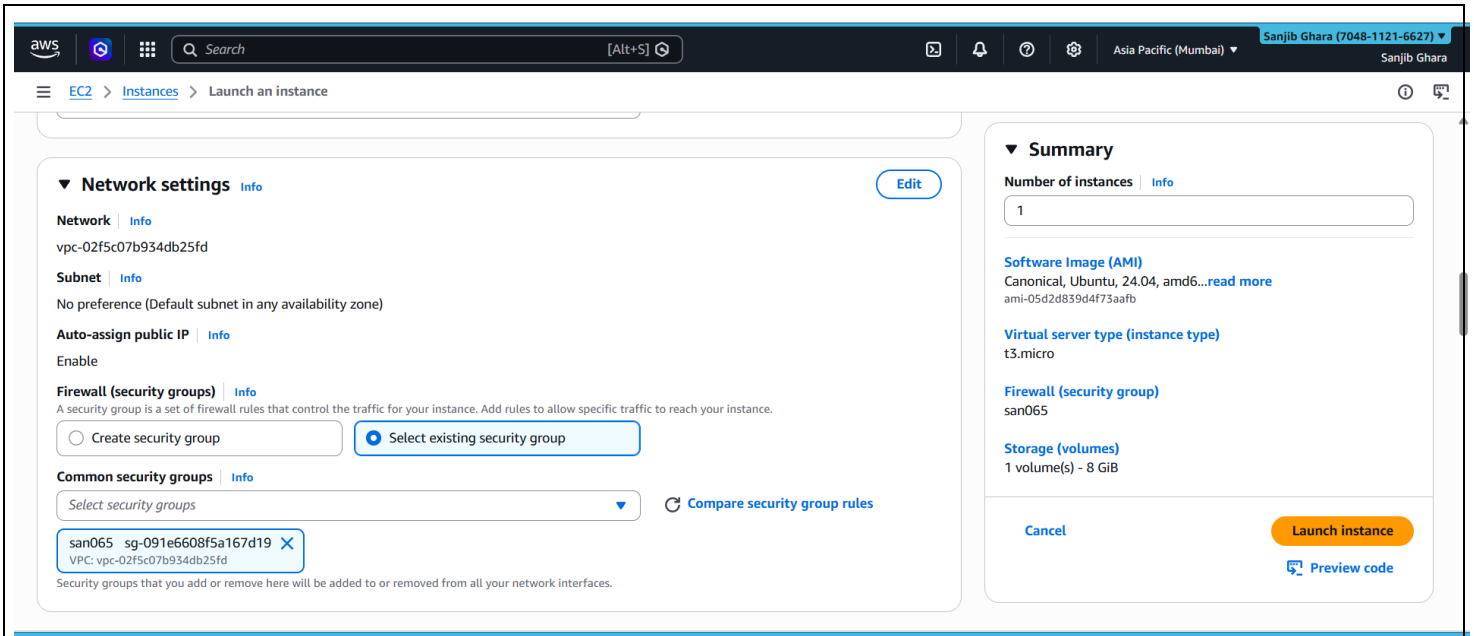
3. Enter:
   - instance name: san-asg14
   - AMI: ubuntu
     - Select instance type: **t3.micro**
     - **Create key pair(skip if already present previous one**

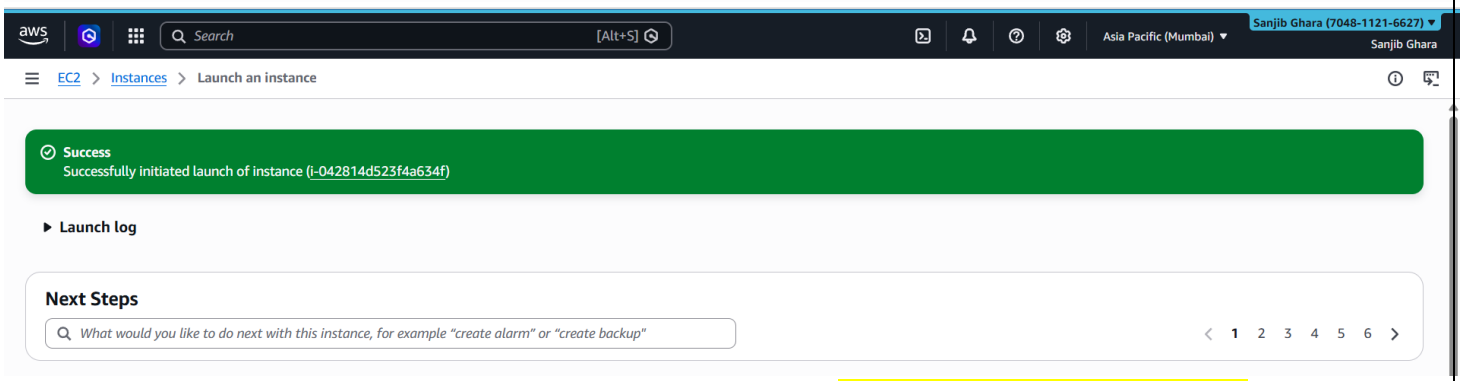
In my case I use the previous one key:webkey1)



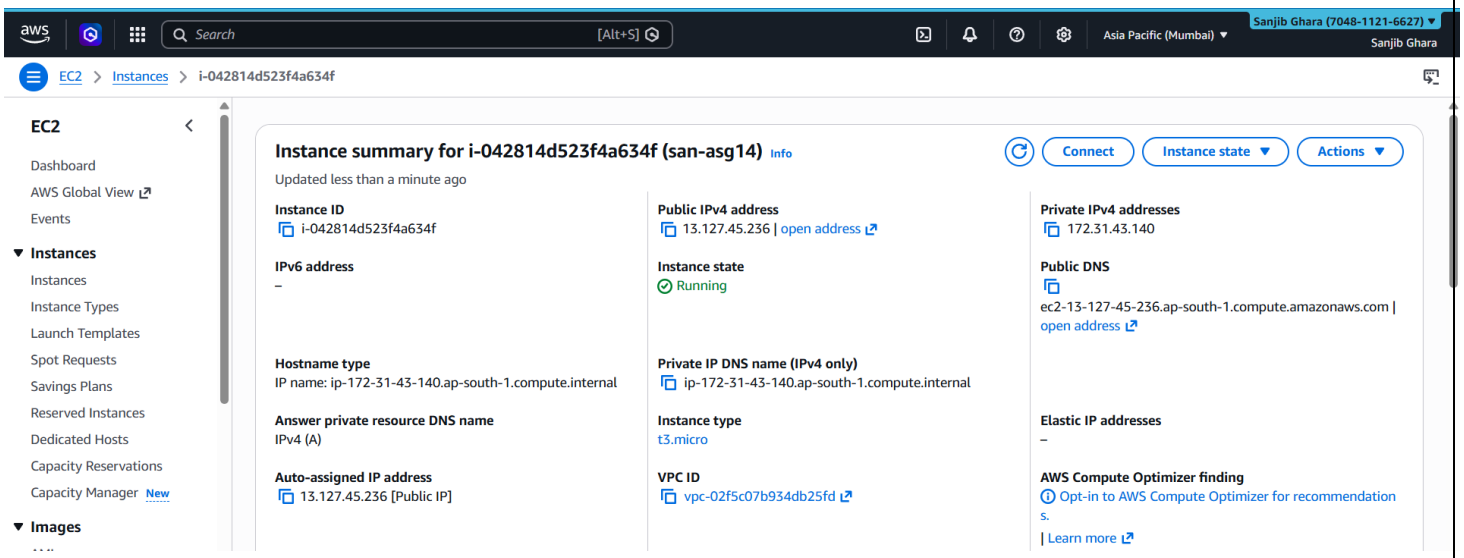


- **Network Setting:** firewall settings (choose select **existing sequrity group)**
- **Create a security group(if already present then use that)**
  In my ,use my previous one.
- **Then press launch instance.**



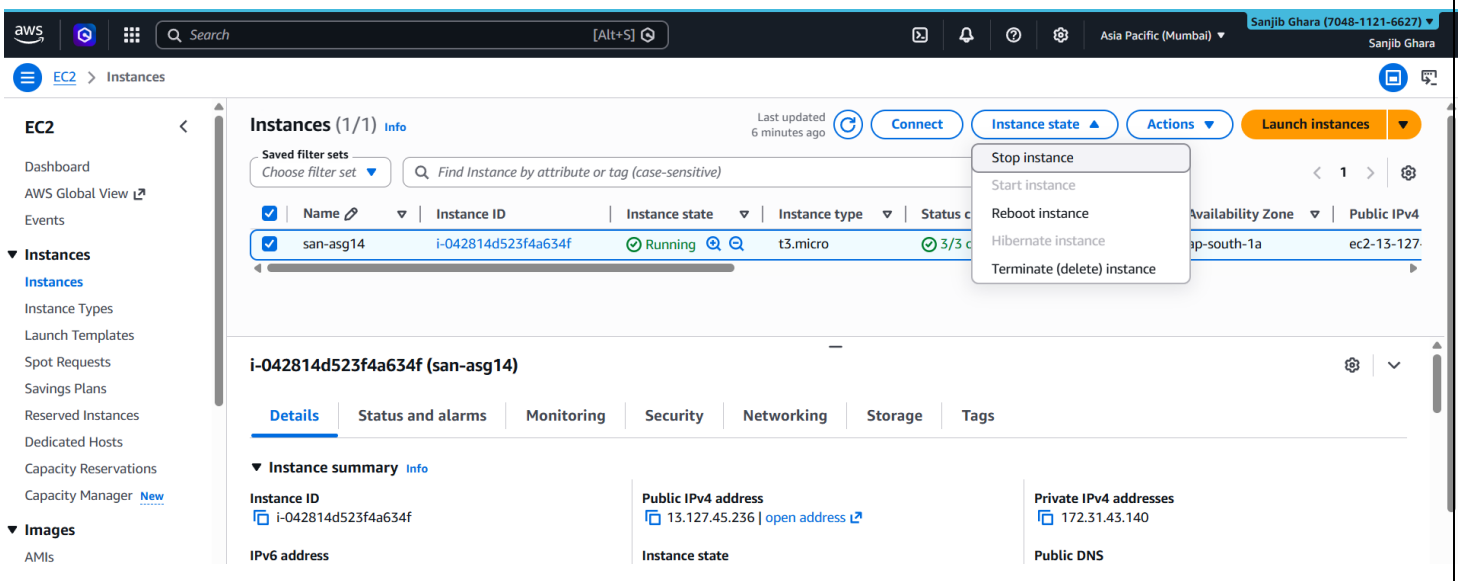


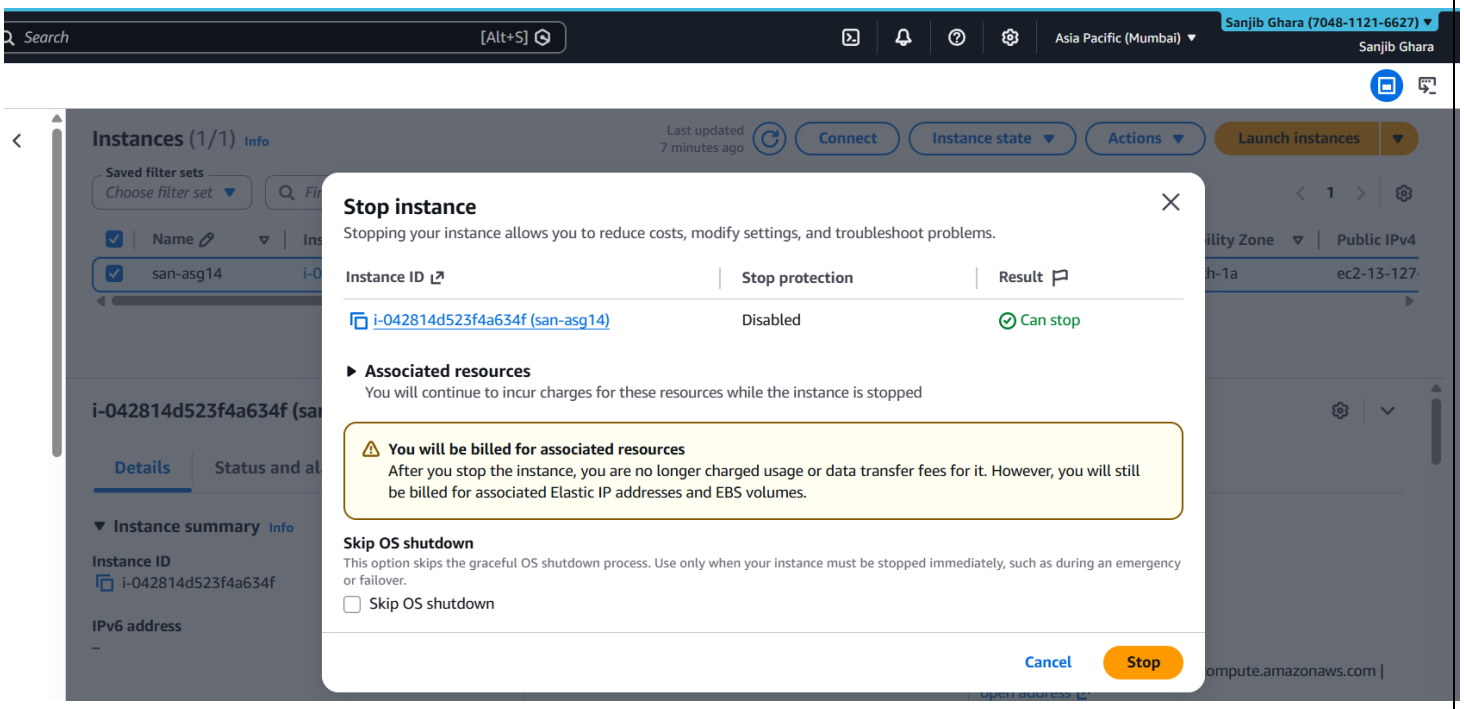
STEP2: Before forcefully stopping the instance the **public ip: 13.127.45.236**
And private ip: 172.31.43.140



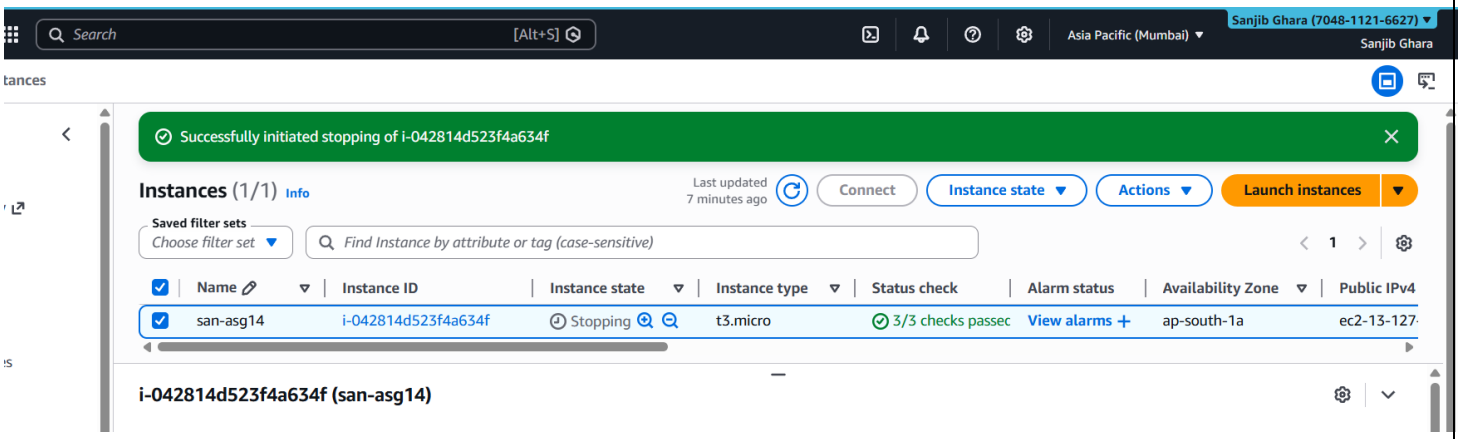
STEP 3:-Stop instance

- Select the instance
- Click on **Action**.
- After that click on **Stop instance**
- **Then** click on **Stop**.





➢ The instance is Successfully stopped.



STEP 4: CHECKINP PUBLIC AND PRIVATE IP'S

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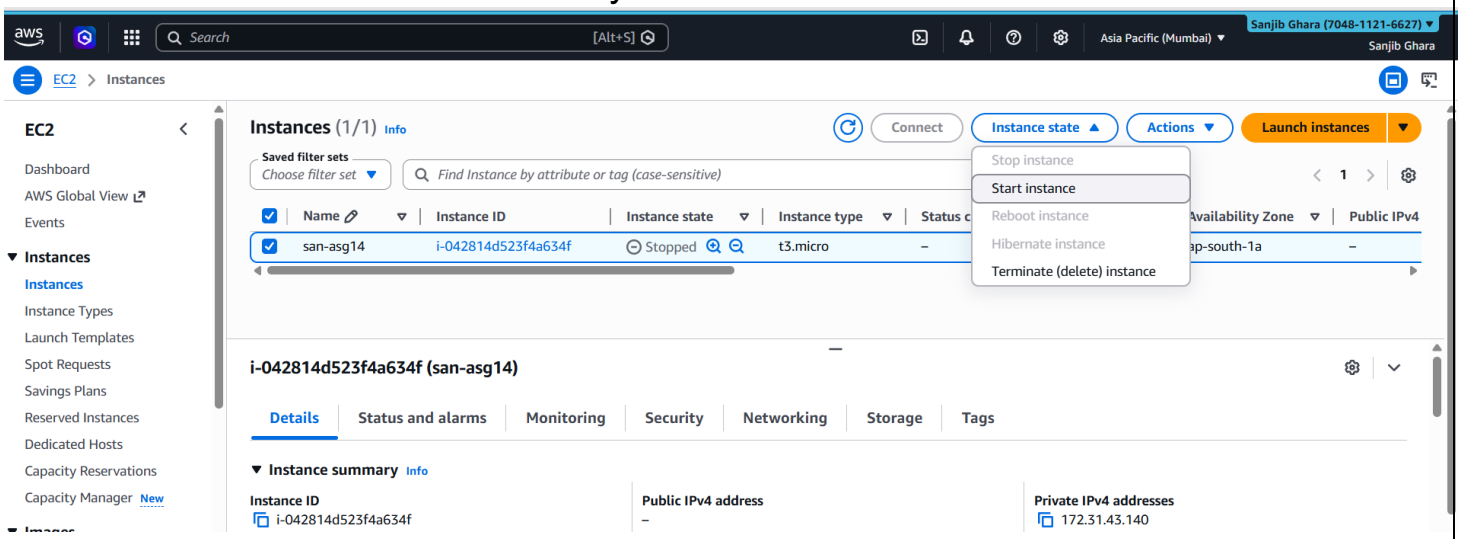
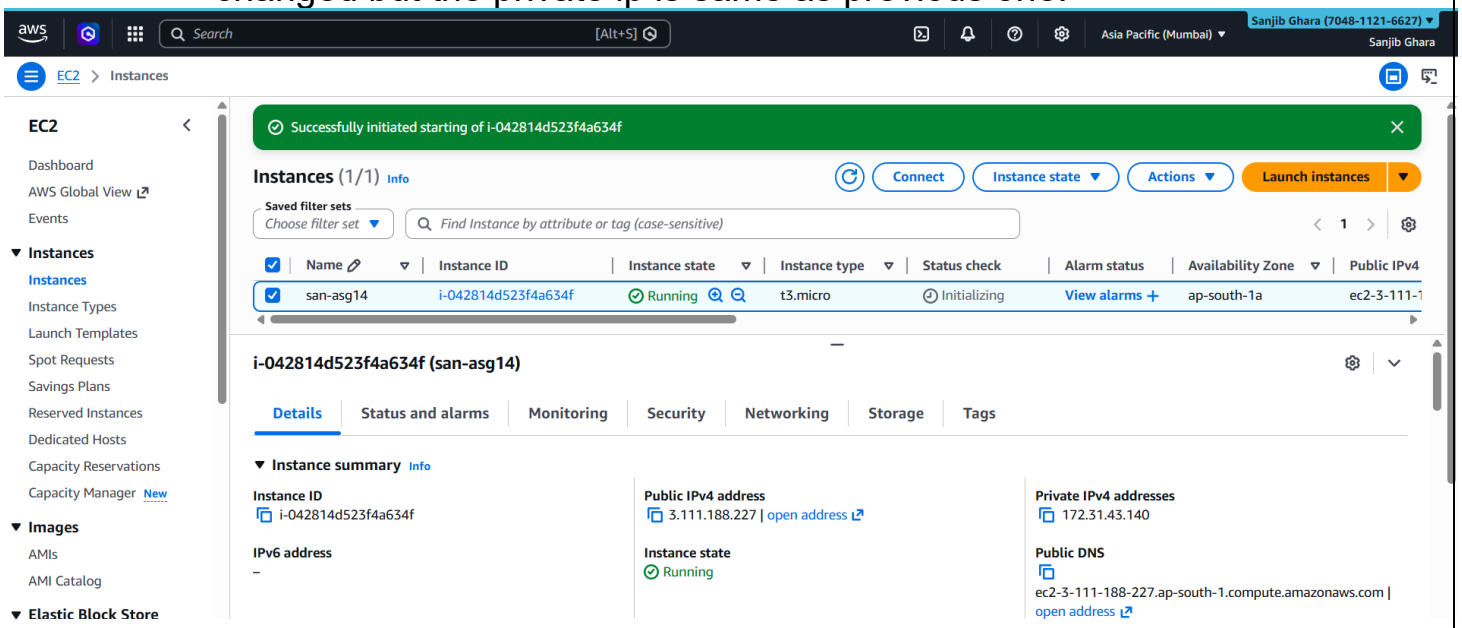
- After stopping, now we restart the instance by click on “START INSTANCE”
- The instance is successfully started.

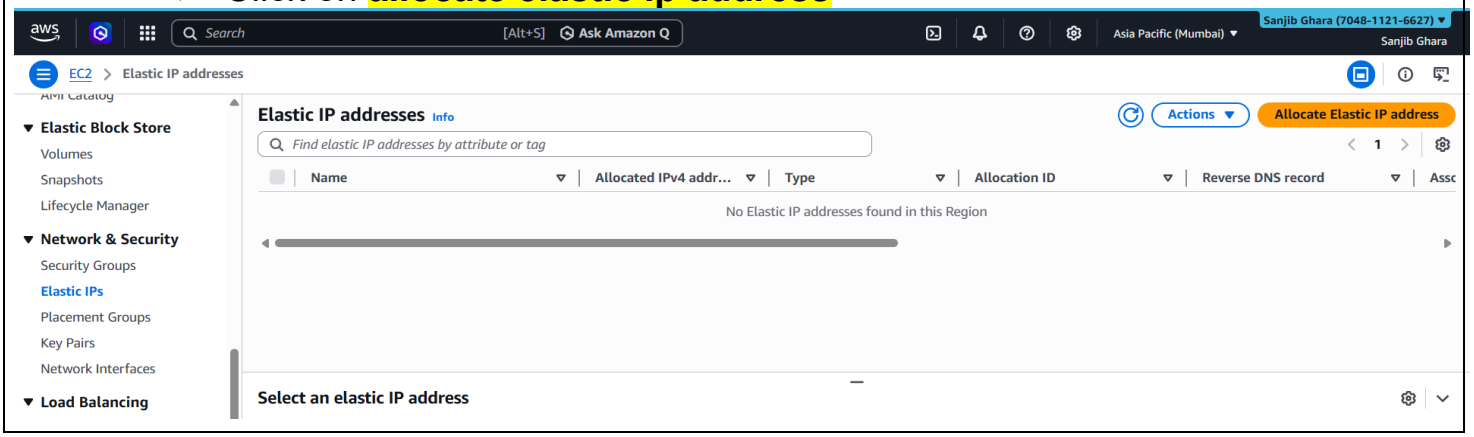


- **Here** we can see that after restarting the public ip: **3.111.188.227 is** changed but the private ip is same as previous one.

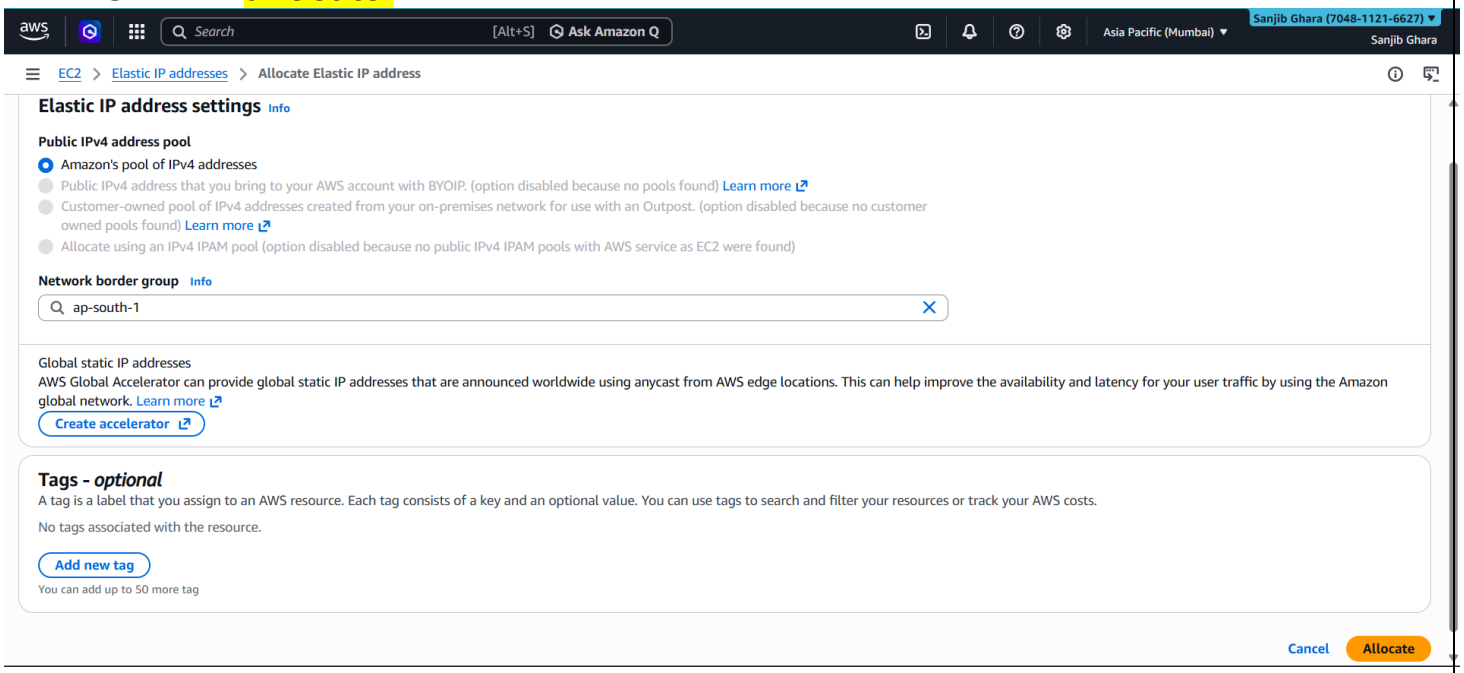


STEP 5: CREATE ELASTIC IP

- Goto -> EC2 (elastic ip)
- Click on **allocate elastic ip address**



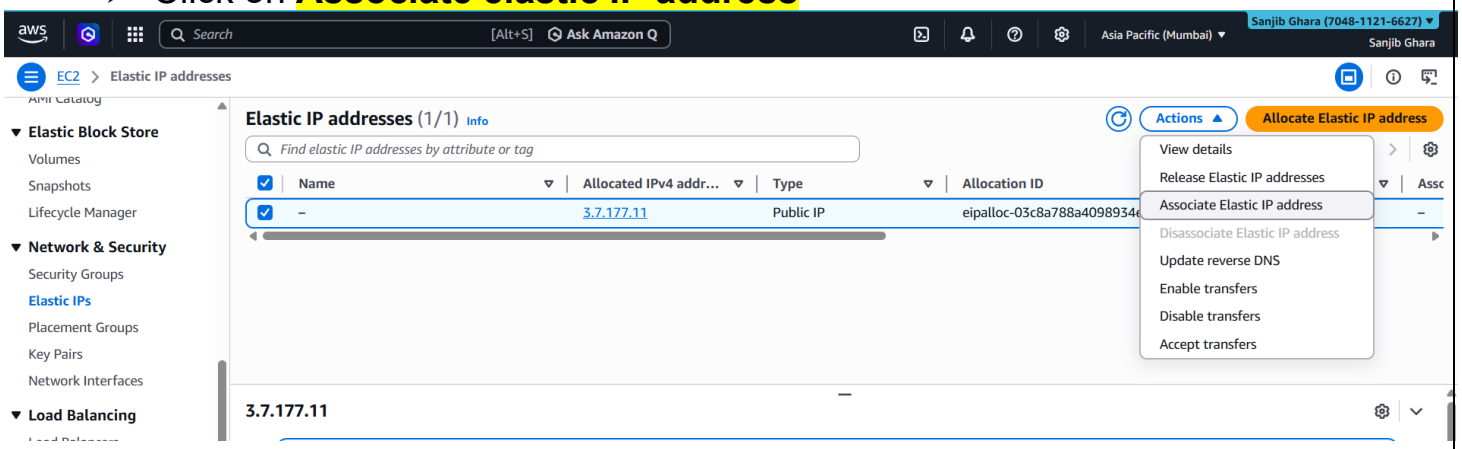
- After that remain all the settings same
- Click on **allocate.**



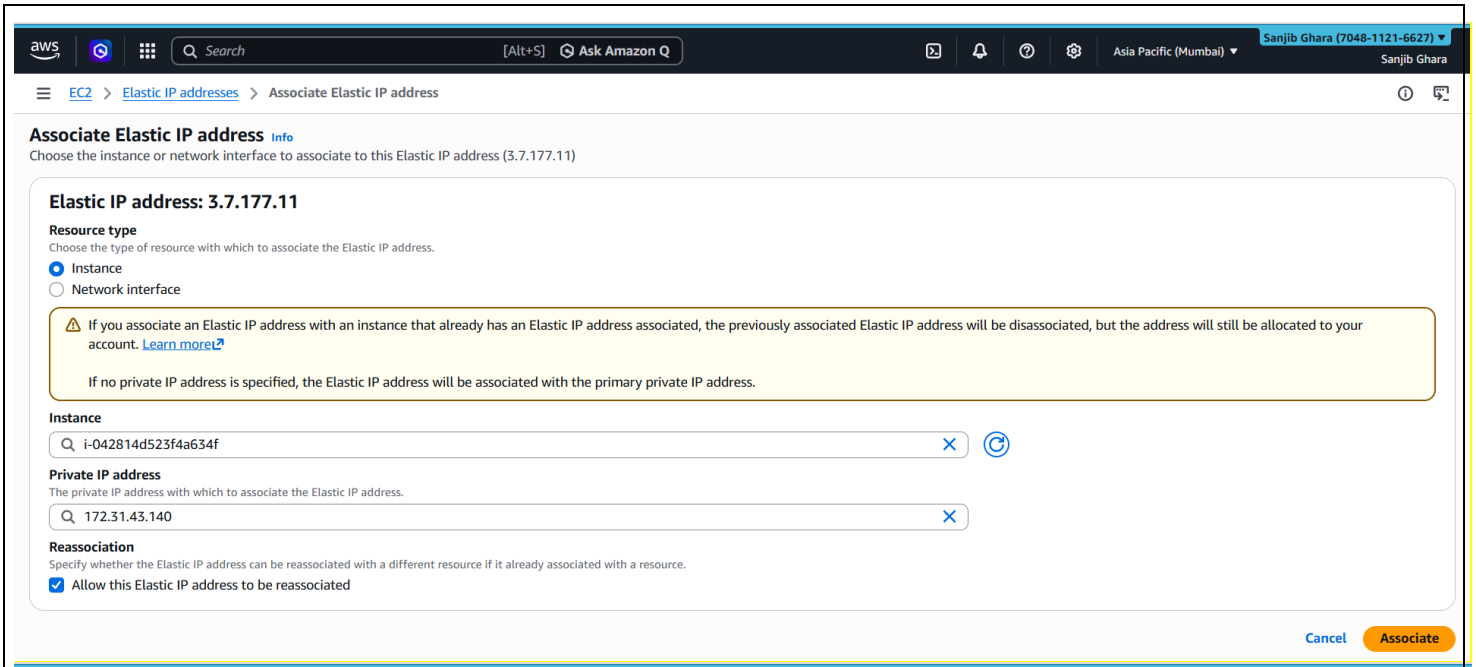
- Successfully created the elastic ip.

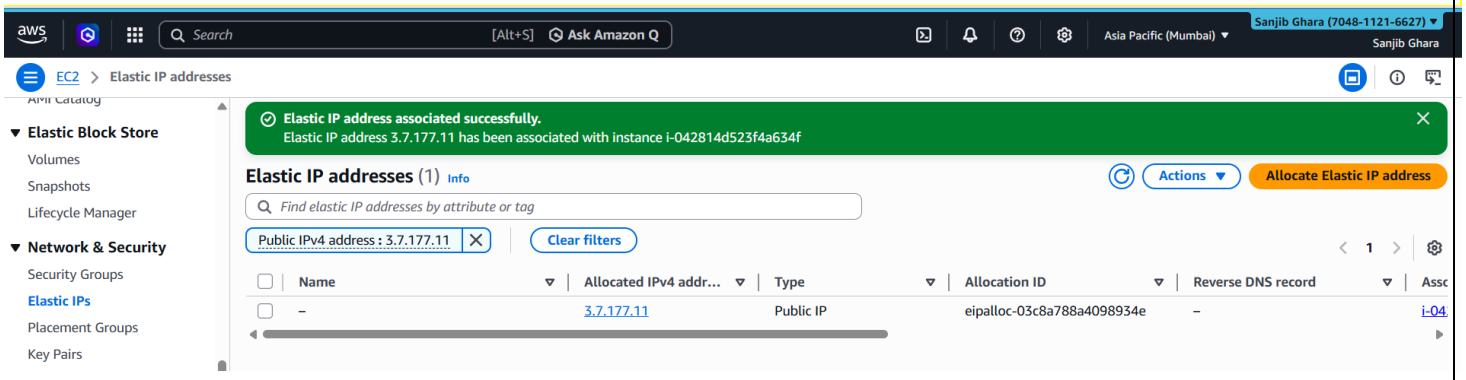
STEP 6: ASSOCIATE ELASTIC IP WITH MY INSTANCE

- SELECT the elastic ip
- Click on **Action.**
- Click on **Associate elastic IP address**



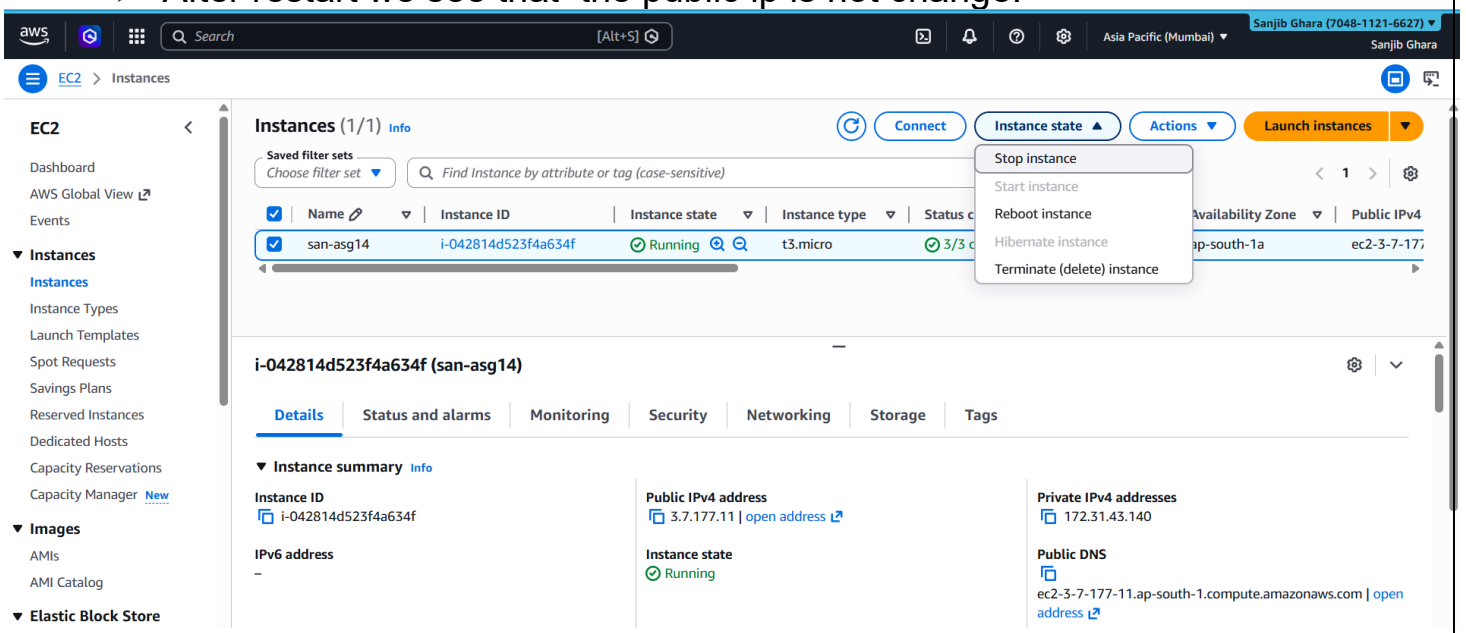
- Then **Resource type** -> instance
- Select Instance -> running instance
- Select private ip address ->in my case(**172.31.43.140)**
- Then tick the check box
- Finally press **Associate.**

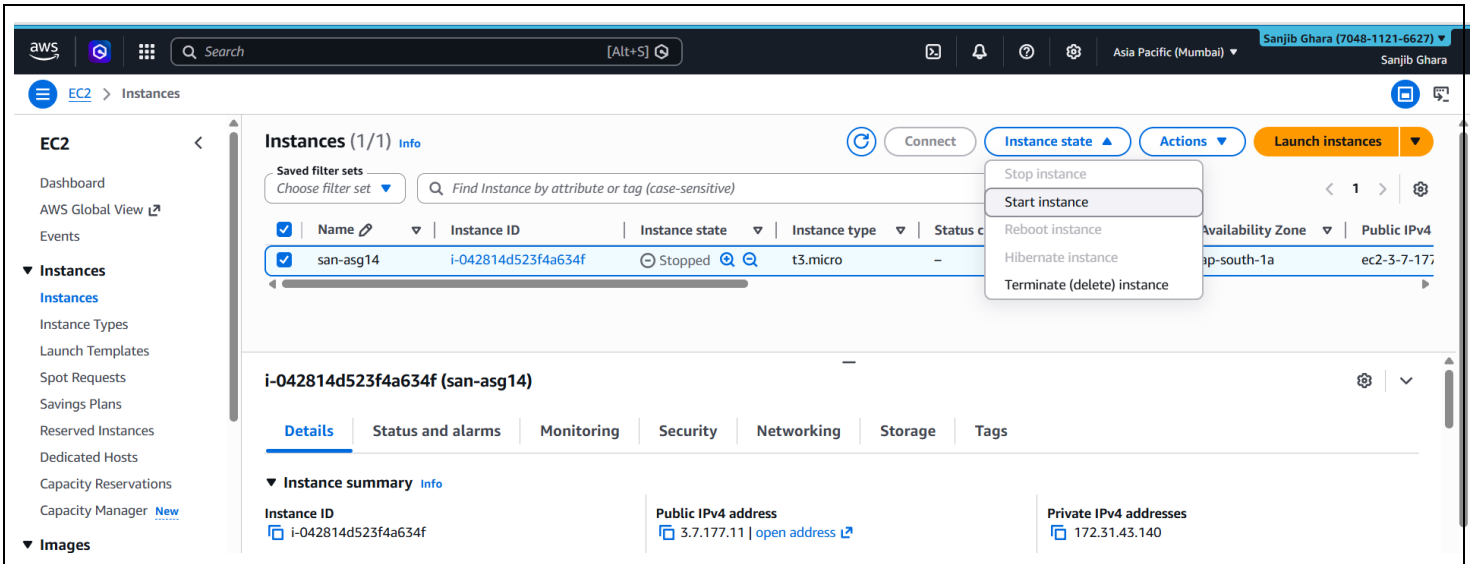


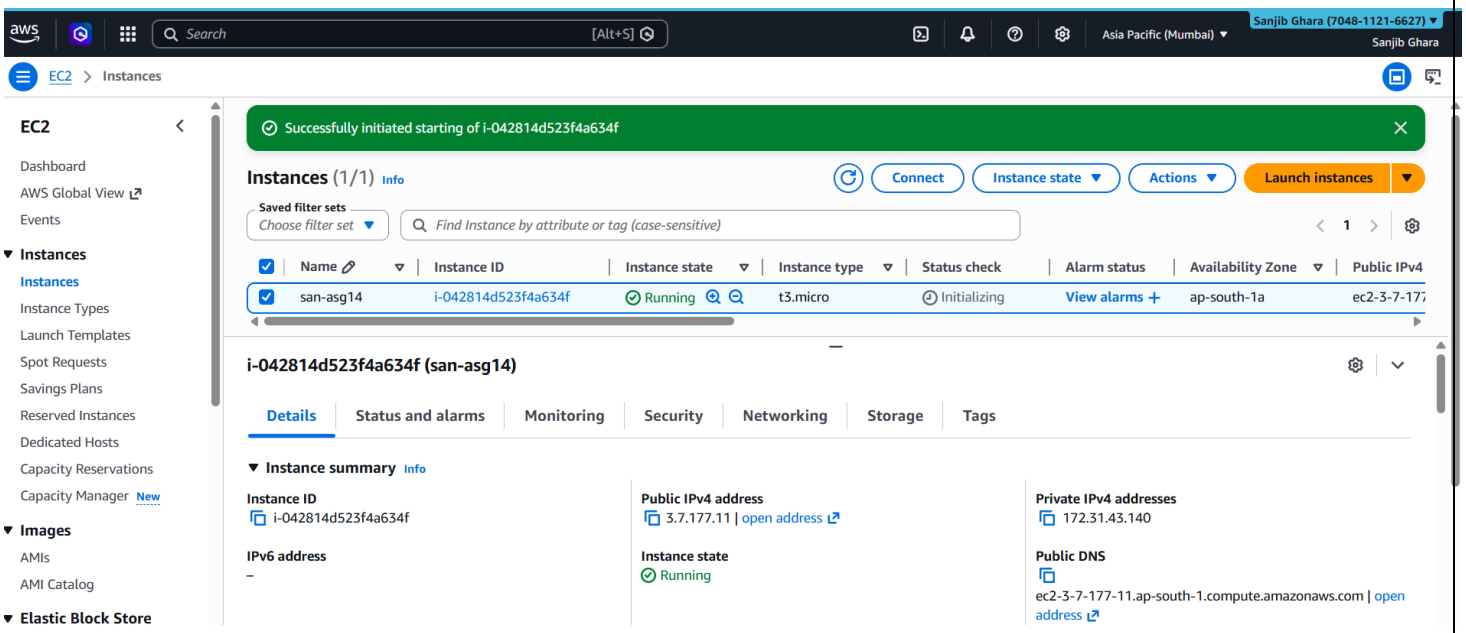


STEP 7: AGAIN CHECKING THE IP

- Again we need to stop the instance
- Then restart
- After restart we see that the public ip is not change.





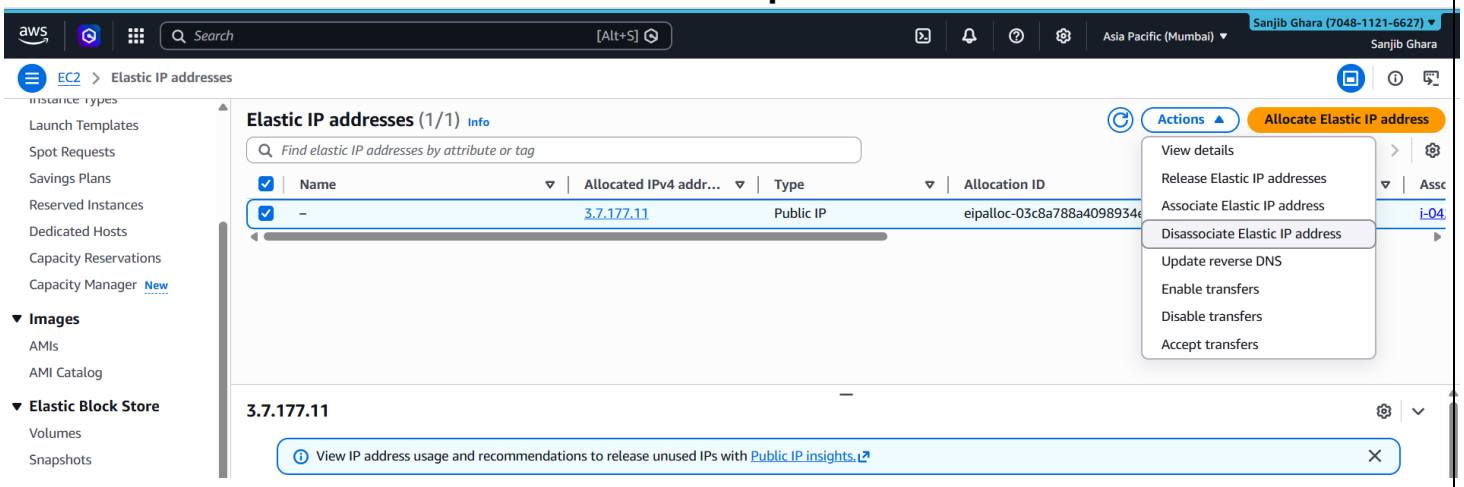


STEP 8:After before result

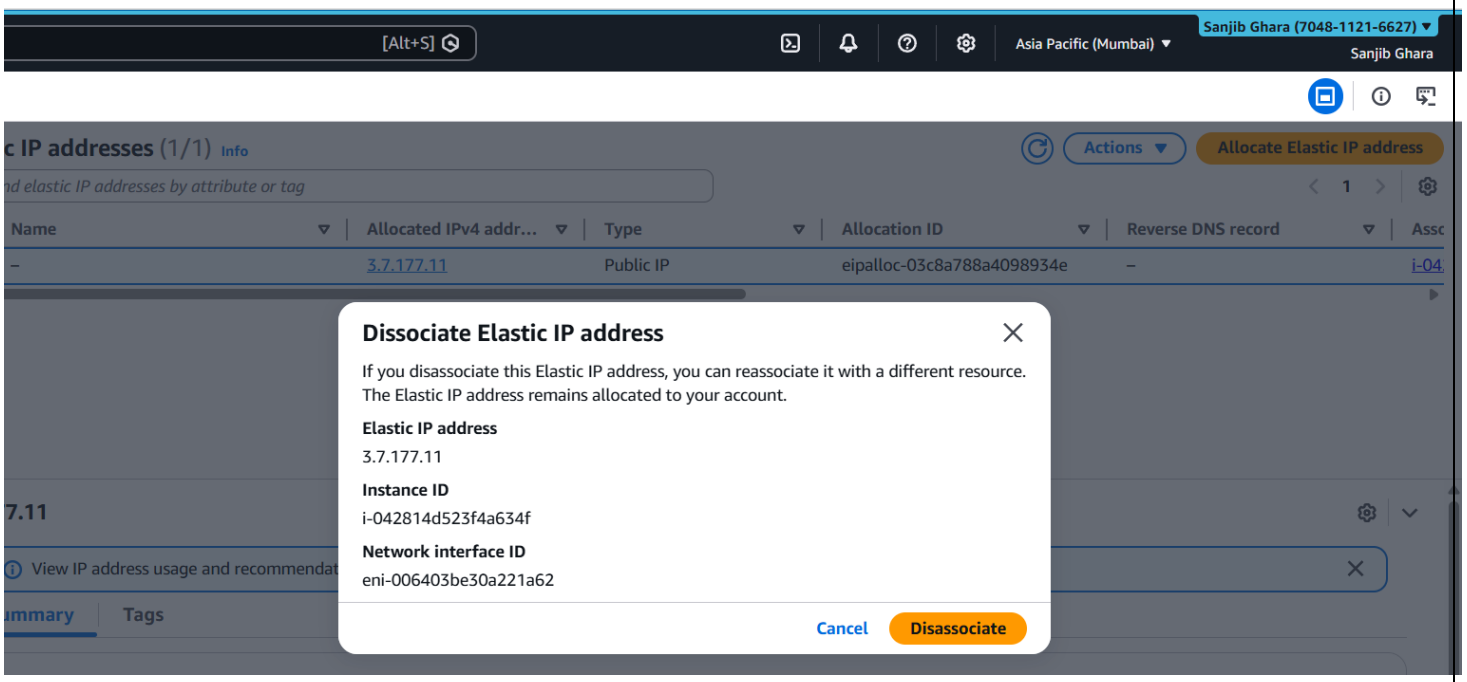
```
Before Elastic Ip assign :        public ip          private ip

  before stop instance          13.127.45.236       172.31.43.140
  after stopping and restart    3.111.188.227       172.31.43.140

after elastic ip assigning:

  before stop instance           3.7.177.11         172.31.43.140

  after stopping and restart      3.7.177.11        172.31.43.140
```

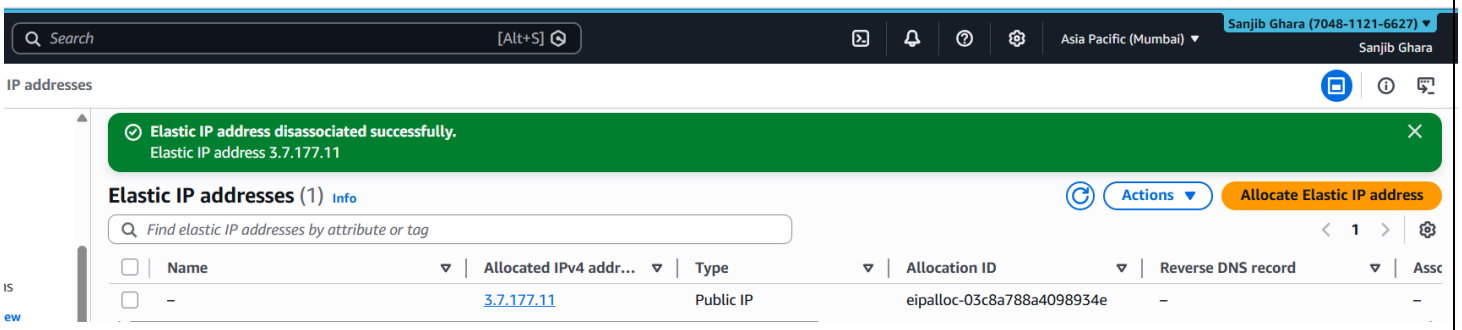
STEP 9: DELETE THE ELASTIC IP

- GOTO -> **elastic ip(EC2)**
- AT first select ip
- Then click on -> **Action.**
- Then click on **-> disassociate elastic ip .**

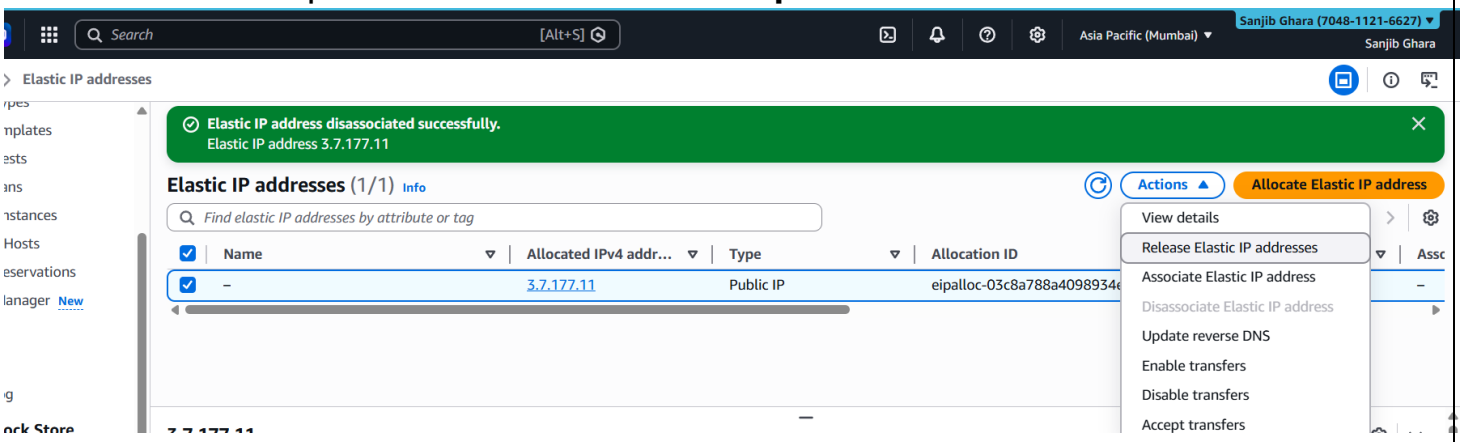


- **Then** select **Dissociate**.

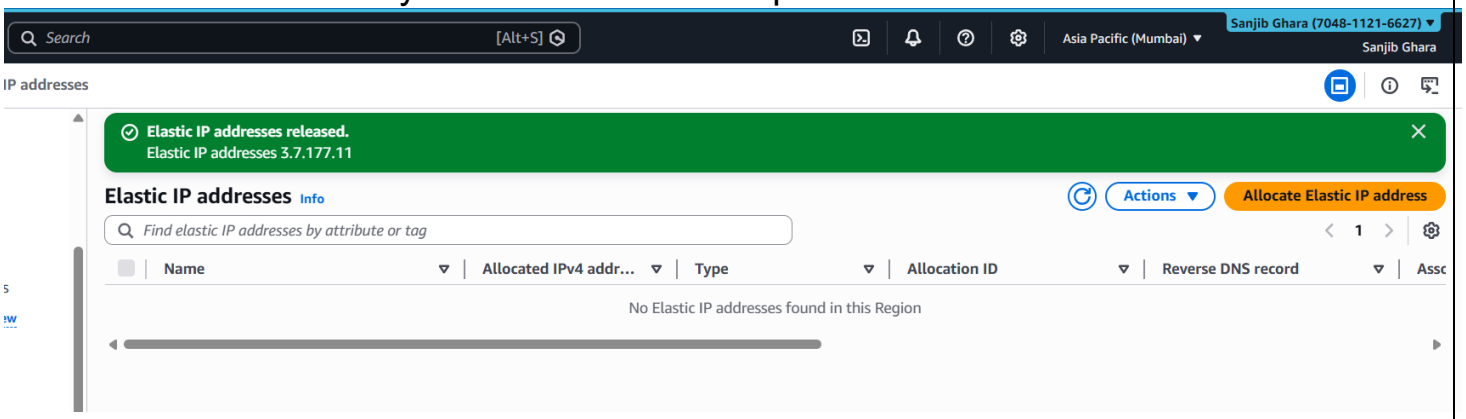




➢ Then press on -> **release elastic ip**



➢ Successfully released the elastic ip.



Conclusion

Elastic IP in **Amazon EC2** provides a **static public IP address** that can be associated with an instance. It ensures that the IP address of the server **remains constant even after stopping or restarting the instance**. By allocating and associating an Elastic IP, users can maintain reliable connectivity and easily remap the address to another instance in case of failure, improving **availability and flexibility** in cloud infrastructure.